

CHRISTMAS ISLAND, Christmas Island

WMO: 969950

Lat: 10.4516S

Lon: 105.6890E

Elev: 262

StdP: 98.22

Time Zone: 7.00 (E07)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.9	21.4	16.3	12.0	23.3	17.2	12.7	23.4	9.1	24.0	8.5	24.1	2.6	130	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	3.7	28.9	25.0	28.3	24.6	28.1	24.5	26.2	27.3	25.7	27.1	25.4	26.9	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	22.0	26.7	25.2	21.0	26.4	25.0	20.7	26.3	83.1	27.0	80.9	27.5	79.4	26.9	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.8	7.4	DB	19.5	29.9	0.8	1.1	18.9	30.7	18.5	31.4	18.0	32.0	17.4	32.8	(4)
(5)			WB	17.5	27.0	1.1	1.3	16.7	28.0	16.1	28.8	15.5	29.5	14.6	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	25.2	25.3	25.6	25.9	25.7	25.1	24.2	23.9	24.1	24.6	24.9	25.1	(6)	
	DBStd	1.02	0.86	0.92	0.89	0.86	0.84	0.76	0.79	0.79	0.62	0.63	0.81	0.90	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	5462	472	428	483	476	488	454	441	431	422	452	448	467	(10)	
	CDD18.3	2421	213	195	224	226	229	204	183	173	172	194	198	208	(11)	
	CDH23.3	13397	1342	1251	1554	1728	1696	1176	646	477	479	777	1040	1231	(12)	
(13) Wind	CDH26.7	827	126	122	172	150	96	22	4	3	2	10	37	83	(13)	
	WSAvg	4.4	2.9	3.0	3.0	3.9	4.7	5.1	5.6	5.8	5.9	5.4	4.5	3.2	(14)	
	PrecAvg	1848	306	341	296	212	171	176	77	45	28	36	156	206	(15)	
	PrecMax	2954	659	811	697	570	529	708	356	164	107	118	526	759	(16)	
	PrecMin	1067	46	60	36	19	14	9	4	2	1	0	4	12	(17)	
	PrecStd	574	149	194	158	146	122	190	79	45	23	38	148	178	(18)	
															(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.1	29.1	29.2	29.1	28.7	27.9	27.2	27.0	27.0	27.5	28.2	29.0	(19)	
		MCWB	24.7	24.9	25.6	25.3	25.4	24.5	24.1	23.9	23.2	23.6	23.7	24.3	(20)	
	2%	DB	28.2	28.4	28.7	28.3	28.1	27.2	26.3	26.1	26.2	27.0	27.6	28.1	(21)	
		MCWB	24.1	24.6	25.3	24.9	24.9	23.9	23.2	22.6	22.6	23.2	23.6	24.0	(22)	
	5%	DB	27.9	28.0	28.2	28.0	27.6	26.9	26.0	25.8	25.9	26.3	27.1	27.6	(23)	
		MCWB	24.0	24.3	24.8	24.7	24.6	23.8	22.9	22.3	22.4	22.9	23.4	23.8	(24)	
	10%	DB	27.3	27.4	27.8	27.6	27.1	26.4	25.6	25.2	25.2	26.0	26.6	27.1	(25)	
		MCWB	23.8	24.2	24.7	24.6	24.3	23.6	22.8	22.1	22.3	22.8	23.2	23.6	(26)	
	0.4%	WB	26.1	25.7	28.0	26.4	27.0	25.9	25.8	25.9	25.2	25.1	25.6	25.7	(27)	
		MCDB	27.7	27.4	28.3	27.7	27.7	26.5	26.1	26.0	25.5	26.0	26.6	27.0	(28)	
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	25.3	25.3	25.8	25.7	25.9	25.1	24.8	24.5	24.2	24.5	24.8	25.2	(29)	
		MCDB	27.0	27.1	27.5	27.3	27.1	26.2	25.6	25.2	25.1	25.6	26.1	26.7	(30)	
	5%	WB	24.9	25.0	25.4	25.5	25.4	24.7	24.2	23.5	23.5	24.1	24.5	24.7	(31)	
		MCDB	26.7	26.9	27.1	27.0	26.7	25.9	25.1	24.7	24.8	25.3	25.8	26.3	(32)	
	10%	WB	24.6	24.7	25.1	25.2	25.0	24.3	23.6	22.8	23.0	23.6	24.1	24.5	(33)	
		MCDB	26.3	26.5	26.8	26.7	26.4	25.6	24.8	24.3	24.4	25.1	25.5	26.1	(34)	
	5% DB	MDBR	4.5	4.3	4.4	3.7	3.2	2.8	2.9	3.1	3.1	3.3	3.6	4.4	(35)	
		MCDBR	5.0	4.9	4.8	3.9	3.4	3.0	3.0	3.3	3.4	3.6	4.0	4.8	(36)	
		MCWBR	2.3	2.2	2.6	2.2	2.0	1.8	1.9	2.0	1.9	1.8	1.9	2.2	(37)	
		MCWBR	4.4	4.2	4.3	3.7	3.2	2.8	2.8	3.0	3.1	3.2	3.5	4.2	(38)	
(39) Clear-Sky Solar Irradiance	taub		0.390	0.399	0.397	0.404	0.412	0.407	0.397	0.394	0.416	0.449	0.450	0.406	(40)	
		taud	2.572	2.556	2.564	2.541	2.504	2.504	2.506	2.490	2.411	2.310	2.340	2.514	(41)	
	Ebn at Noon		951	938	923	886	849	837	854	883	892	881	889	934	(42)	
		Edh at Noon	107	108	105	103	101	99	100	107	121	137	134	113	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.24	6.17	5.86	5.26	5.02	4.68	5.02	5.72	6.22	6.56	6.30	6.28	(44)	
	RadStd		0.66	0.61	0.50	0.47	0.36	0.45	0.35	0.16	0.45	0.55	0.59	0.66	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.28	+0.52	N/A	N/A	+0.44	+0.79	N/A	N/A	N/A	N/A	(46)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page